

Problem-Solution Fit

Date	30 June 2026
Team ID	XXXXXX
Project Name	Flood Prediction System
Maximum Marks	5 Marks

Problem-Solution Fit:

1. CUSTOMER SEGMENT(S) [CS] - Disaster Management Authorities - Government Agencies - Weather Monitoring Departments - Residents in Flood-Prone Areas	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] - Limited real-time information - Limited access to advanced prediction systems - Dependence on manual monitoring - Resource constraints in disaster management	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Current Solutions - Manual weather monitoring - Traditional forecasting methods - Government weather alerts Pros - Easily accessible - Based on official weather data Cons - Limited prediction accuracy - Delayed alerts - Manual analysis required
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] - Difficulty predicting floods accurately - Delayed flood warnings - Property and infrastructure damage - Frequent during heavy rainfall seasons	9. PROBLEM ROOT / CAUSE [RC] - Climate change - Heavy and unpredictable rainfall - Poor drainage systems - Rapid urbanization - Lack of early prediction systems	7. BEHAVIOR + ITS INTENSITY [BE] - Regularly monitor weather forecasts - Check rainfall reports during monsoon - Respond quickly to flood alerts - High urgency during heavy rainfall

3. TRIGGERS TO ACT [TR]

- Heavy rainfall
- Rising water levels
- Extreme weather forecasts
- Monsoon season alerts

4. EMOTIONS BEFORE / AFTER [EM]

Before

- Worried
- Uncertain
- Anxious

After

- Confident
- Prepared
- Safer

10. YOUR SOLUTION [SL]

Develop a Machine Learning-based Flood Prediction System that analyzes weather parameters such as cloud cover and rainfall to predict flood risk. The system provides quick and reliable predictions through a Flask web application, helping disaster management authorities and communities take preventive action and improve disaster preparedness.

8. CHANNELS OF BEHAVIOR [CH]

ONLINE

- Flood Prediction Web Application
- Government Weather Websites
- Mobile Applications

OFFLINE

- Disaster Management Centers
- Local Authorities
- Emergency Response Teams